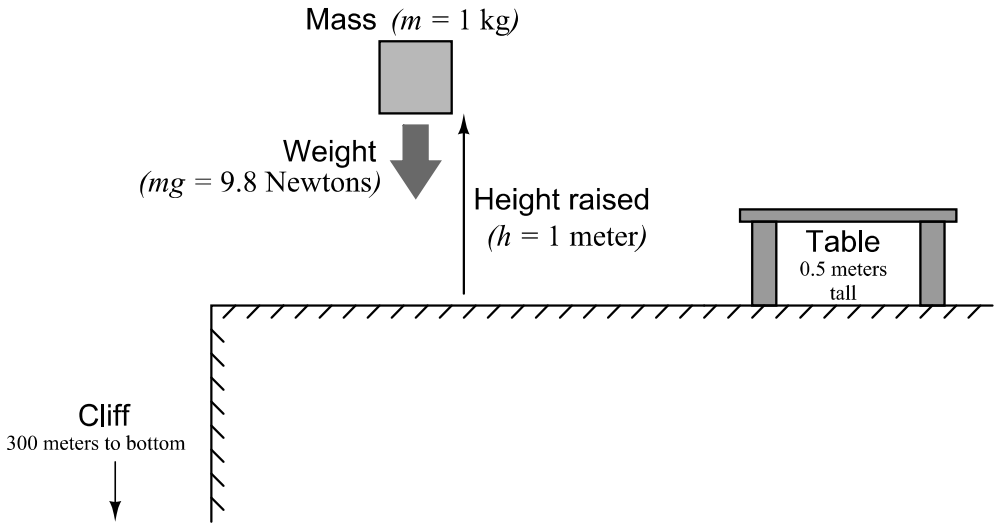
If we make use of non-mechanical means to move electric charge from one location to another, the result is no different. Moving attracting charges apart from one another means doing *work* (a force exerted over a parallel distance) and storing potential energy in that physical tension. When we use chemical reactions to move electrons from one metal plate to another in a solution, or when we spin a generator and electro-magnetically motivate electrons to seek other locations, we impart potential energy to those electrons. We could express this potential energy in the same unit as we do for mechanical systems (the *Joule*). However, it is actually more useful to express the potential energy in an electric system in terms of how many joules are available per a specific quantity of electric charge (a certain number of electrons). This measure of *specific* potential energy is simply called *electric potential* or *voltage*, and we measure it in units of *Volts*, in honor of the Italian physicist Alessandro Volta, inventor of the first electrochemical battery.

$$1 \text{ Volt} = \frac{1 \text{ Joule of potential energy}}{1 \text{ Coulomb of electric charge}}$$

In other words, if we forced 1 Coulomb's worth of electrons (6.24×10^{18} of them, to be exact) away from a positively-charged place, and did one Joule's worth of work in the process, we would have generated one Volt of electric potential.

Electric potential (voltage) and potential energy share a common, yet confusing property: both quantities are fundamentally *relative* between two physical locations. There is really no such thing as specifying a quantity of potential energy at a single location. The amount of potential energy in any system is always relative between two different points. If I lift a mass off the ground, I can specify its potential energy, *but only in relation to its former position on the ground.* The amount of energy that mass is potentially capable of releasing by free-fall depends on how far it could possibly fall. To illustrate, we will perform some "thought experiments" where we lift a 1 kilogram mass 1 meter off the ground and then drop that mass to assess its energy. That 1-kilo mass weighs 9.81 Newtons on Earth, and the distance lifted was 1 meter, so the potential energy stored in the mass is 9.81 joules, right? Consider the following scenario:



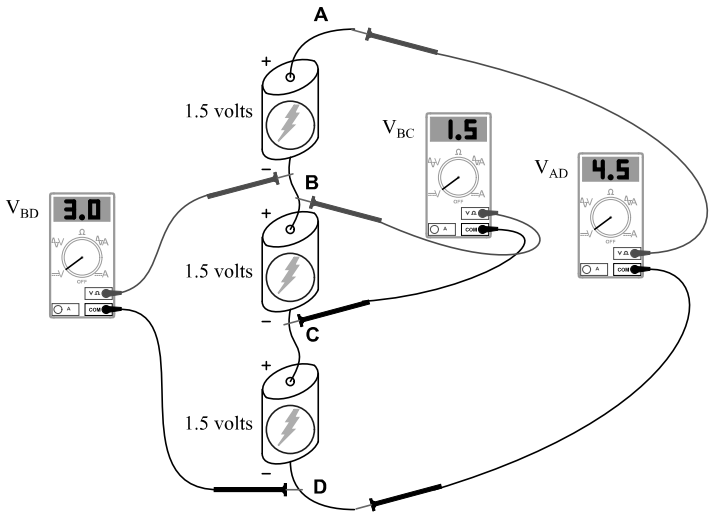
If we drop the mass over the spot we first lifted it from, it will release all the potential energy we invested in it: 9.81 joules. From this example it seems very clear that the mass possessed 9.81 joules of potential energy when lifted. However, the situation becomes more complicated if we carry the mass over to the table and release it there. Since the mass can now only fall *half* a meter, it will only release 4.9 joules of energy in the process. How much potential energy did the mass have while suspended above that table? If you argue that the mass possessed a full 9.81 joules of potential energy at that original height, then why can't we realize that full potential when dropped over the table? Can we really say that an object possess potential energy if we cannot realize that potential?

Next, imagine lifting the 1 kg mass 1 meter off the ground, and then carrying it over the edge of a cliff with a 300 meter drop. If we were to release the mass here, it would fall a full 301 meters before hitting the ground. This represents *2.95 kilojoules (kJ)* of energy! How can we possibly argue that the mass had a potential energy of 9.81 joules when lifted one meter off the ground, and then without imparting any more energy to that mass we suddenly are able to realize over 300 times as much energy?

The answer to this paradox is that potential energy is always *relative* between the object's initial height and its height after falling. It is technically incorrect to say that the mass possesses 9.81 joules of potential energy when lifted 1 meter off the ground, as though potential energy were an absolute

quantity. Instead, it is more accurate to say that the mass *gained* 9.81 joules of potential energy after being lifted 1 meter off the ground, implying that the mass could have possessed potential energy (with reference to the cliff's bottom) even while sitting on the ground (above the cliff).

Likewise, electrical voltage (as an expression of potential energy) is a quantity relative between two points. Electrical voltage is symbolized by the variable V and measured in the unit of the *Volt*. Just as we must know the mass's position relative to its falling point before we can quantify its potential energy, we must know an electric charge's position relative to its return point before we can quantify its voltage. Consider a series of batteries[1] connected as shown:



The voltage as measured between any two points directly across a single battery will be 1.5 volts:

$V_{AB} = 1.5$ volts

$V_{BC} = 1.5$ volts

$V_{CD} = 1.5$ volts

If, however, we span more than one battery with our voltmeter connections, our voltmeter will register more than 1.5 volts:

$V_{AC} = 3.0$ volts

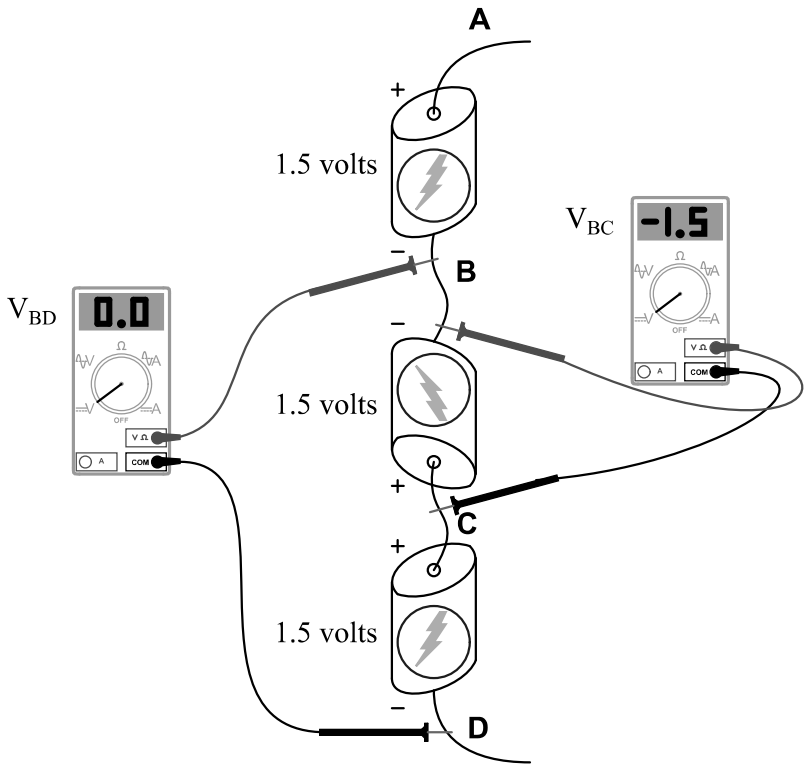
$V_{BD} = 3.0$ volts

$V_{AD} = 4.5$ volts

Thus we see there is no such thing as "voltage" at a single point in a circuit. The concept of voltage has meaning only *between* pairs of points in a circuit, just as the concept of potential energy for a mass has meaning only *between* two physical locations: where the mass is, and where it may potentially fall.

[1] A *battery* is an electrochemical device producing an electrical voltage as the result of a chemical reaction.

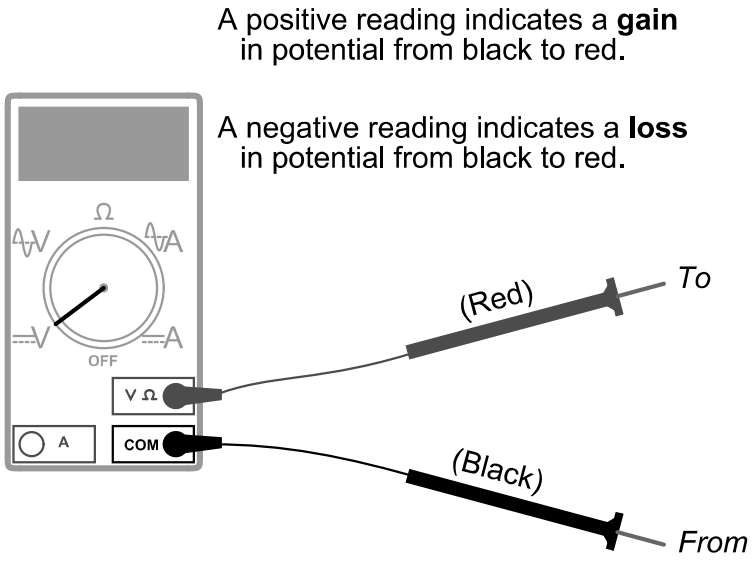
Things get interesting when we connect voltage sources in different configurations. Consider the following example, identical to the previous illustration except the middle battery has been reversed:



Note the "+" and "−" signs next to the ends of the batteries. These signs show the *polarity* of each battery's voltage. Also note how the two voltmeter readings are different from before. Here we see an example of *negative potential* with the middle battery connected in opposition to the other two batteries. While the top and bottom batteries are both "lifting" electric charges to greater potential (going from point **D** to point **A**), the middle battery is decreasing potential from point **C** to point **B**. It's like taking a step forward, then a step back, then another step forward. Or, perhaps more appropriately, like lifting a mass 1.5 meters up, then setting it down 1.5 meters, then lifting it 1.5 meters up again. The first and last steps accumulate potential energy, while the middle step releases potential energy.

This explains why it is important to install multiple batteries the same way into battery-powered devices such as radios and flashlights. The batteries' voltages are supposed to add to a make a larger total required by the device. If one or more batteries are placed backwards, potential will be lost instead of gained, and the device will not receive enough voltage.

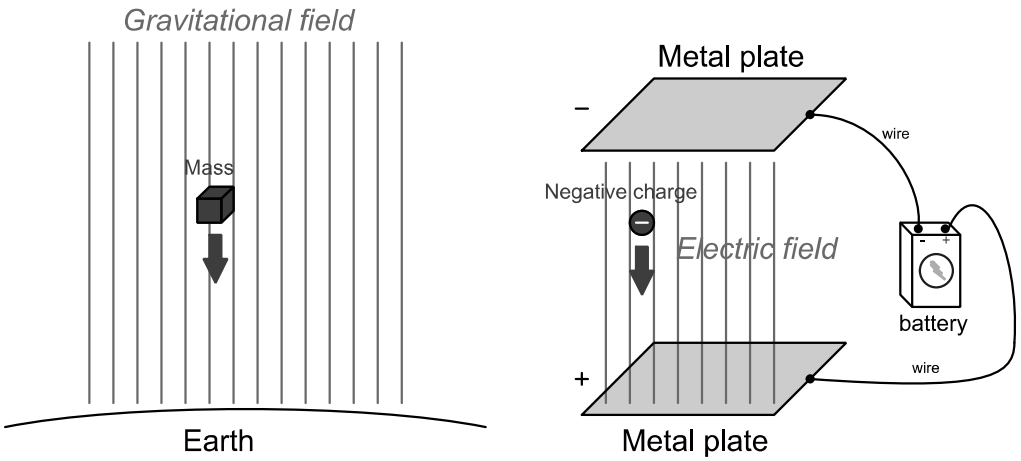
Here we must pay special attention to how we use our voltmeter, since polarity matters. All voltmeters are standardized with two colors for the test leads: red and black. To make sense of the voltmeter's indication, especially the positive or negative *sign* of the indication, we must understand what the red and black test lead colors mean:



Connecting these test leads to different points in a circuit will tell you whether there is potential gain or potential loss from one point (black) to the other point (red).

4.2 Electrical current

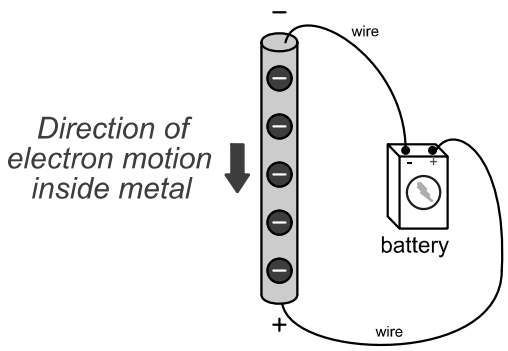
Current is the name we give to the motion of electric charges from a point of high potential to a point of low potential. All we need to form an electric current is a source of potential (voltage) and some electric charges that are free to move between the poles of that potential. For instance, if we connected a battery to two metal plates, we would create an electric field between those plates, analogous to a gravitational field except it only acts on electrically charged objects, while gravity acts on anything with mass. A free charge placed between those plates would "fall" toward one of the plates just as a mass would fall toward a larger mass:



An electric charge will "fall" in an electric field just as a mass will fall in a gravitational field.

Some solid substances, most notably metals, have very mobile electrons. That is, the outer (valence) electrons are very easily dislodged from the parent atoms to drift to and fro throughout the material. In fact, the electrons within metals are so free that physicists sometimes refer to the structure of a metal as atomic nuclei "floating in a sea of electrons". The electrons are almost fluid in their mobility throughout a solid metal object, and this property of metals may be exploited to form definite pathways for electric currents. This is why electrical wires are made of metal such as copper or aluminum: substances with a high degree of electron mobility. Any substance whose electrons are mobile is called a *electrical conductor*, while any substance lacking mobile electrons is called an *electrical insulator*.

If the poles of a voltage source are joined by a conductor, the free electrons within that conductor will drift toward the positive pole (electrons having a negative charge, opposite charges attracting one another). For each electron reaching the positive pole, an electron exits the negative pole of the source to replenish the total number of electrons in the flow:

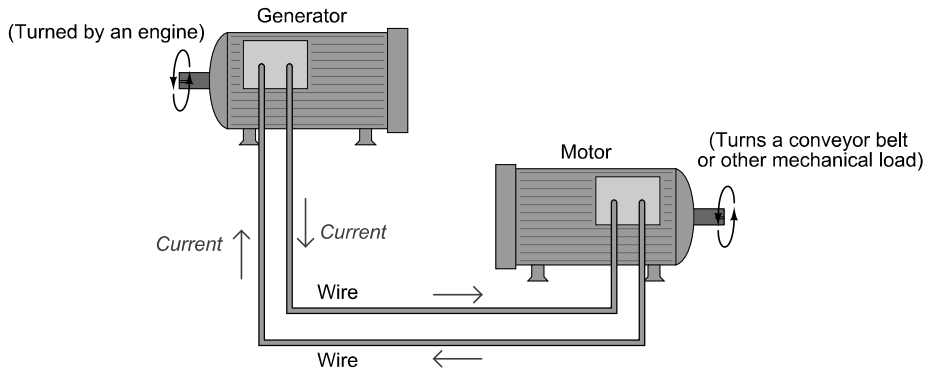


If the source of this voltage is continually replenished by chemical energy, mechanical energy, or some other form of energy, the free electrons will continually loop around this circular path. We call this unbroken path an *electric circuit*. The drifting motion of electrons in a circuit has the same average rate of flow (current) at all points in that circuit, because there is only one pathway for the current. You may think of this like liquid flowing through a circular loop of pipe: since there is only one pathway for the liquid to flow, the rate of flow at all points in that pathway must be the same.

We typically measure the amount of current in a circuit by the unit of *Amperes*, or *Amps* for short (named in honor of the French physicist André Ampère, and symbolize it in formulae using the variable I. One ampere of current is equal to one coulomb of electric charge (6.24×10^{18} electrons) moving past a point in a circuit for every second of time.

Like masses falling toward a source of gravity, these electrons continually "fall" toward the positive pole of a voltage source. After arriving at that source, the energy imparted by that source "lifts" the electrons to a higher potential state where they once again "fall down" to the positive pole through the circuit.

Like rising and falling masses in a gravitational field, these electrons act as carriers of energy within the electric field of the circuit. This is very useful, as we can use them to convey energy from one place to another, using metal wires as conduits for this energy. This is the basic idea behind electric power systems: a source of power (a *generator*) is turned by some mechanical engine (windmill, water turbine, steam engine, etc.), creating an electric potential. This potential is then used to motivate free electrons inside the metal wires to drift in a common direction. The electron drift is conveyed in a circuit through long wires, where they can do useful work at a *load* device such as an electric motor, light bulb, or heater.



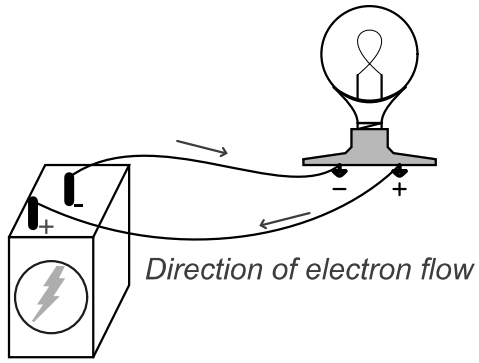
Given the proper metal alloys, the friction that electrons experience within the metal wires may be made very small, allowing nearly all the energy to be expended at the load (motor), with very little wasted along the path (wires). This makes electricity the most efficient means of energy transport known.

The electric currents common in electric power lines may range from hundreds to thousands of amperes. The currents conveyed through power receptacles in your home typically are no greater than 20 amperes. The currents in the small battery-powered circuits you will build are even less: fractions of an ampere. For this reason, we commonly use the metric prefix *milli* (one one-thousandth) to express these small currents. For instance, 10 milliamperes is 0.010 amperes, and 500 milliamperes is one-half of an ampere.

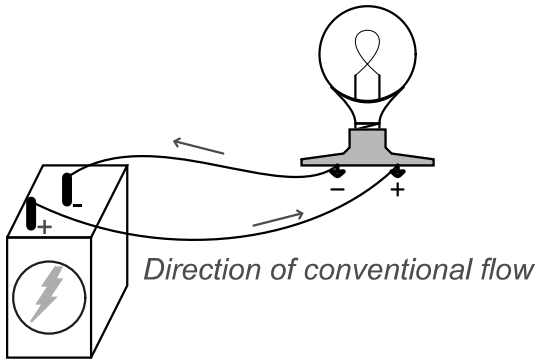
4.2.1 Electron versus conventional flow

When Benjamin Franklin proposed his single-fluid theory of electricity, he defined "positive" and "negative" as the surplus and deficiency of electric charge, respectively. These labels were largely arbitrary, as Mr. Franklin had no means of identifying the actual nature of electric charge carriers with the primitive test equipment and laboratory techniques of his day. As (bad) luck would have it, his hypothesis was precisely opposite of the truth for metallic conductors, where electrons are the dominant charge carrier.

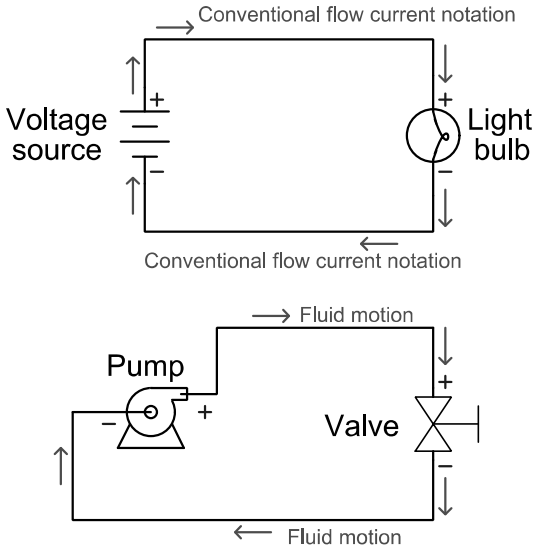
This means that in an electric circuit consisting of a battery and a light bulb, electrons slowly move from the negative side of the battery, through the metal wires, through the light bulb, and on to the positive side of the battery as such:



Unfortunately, scientists and engineers had grown accustomed to Franklin's false hypothesis long before the true nature of electric current in metallic conductors was discovered. Their preferred notation was to show electric current flowing from the positive pole of a source, through the load, returning to the negative pole of the source:

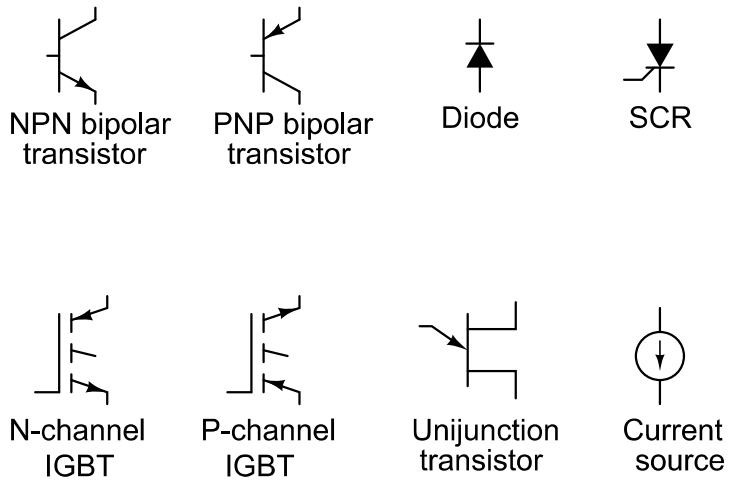


This relationship between voltage polarity marks and conventional flow current tends to make more intuitive sense than electron flow notation, because it is reminiscent of fluid pressure and flow direction:



If we take the "+" sign to represent *more* pressure and the "−" sign to represent *less* pressure, it makes perfect sense that fluid should move from the high-pressure (discharge) port of the pump through the hydraulic "circuit" and back to the low-pressure (suction) port of the pump. It also makes perfect sense that the upstream side of the valve (a fluid restriction) will have a greater pressure than the downstream side of the valve. In other words, conventional flow notation best honors Mr. Franklin's original intent of modeling current as though it were a fluid, even though he was later proven to be mistaken in the case of metallic conductors where electrons are the dominant charge carrier.

This convention was so well-established in the electrical engineering realm that it held sway despite the discovery of electrons. Engineers, who create the symbols used to represent the electronic devices they invent, consistently chose to draw arrows in the direction of conventional flow rather than electron flow. In each of the following symbols, the arrow heads point in the direction that *positive* charge carriers would move (opposite the direction that electrons actually move):



This stands in contrast to electronics technicians, who historically have been taught using electron flow notation. I remember sitting in a technical school classroom being told by my teacher to always imagine the electrons moving *against the arrows* of the devices, and wondering why it mattered.

It is truly a sad situation when the members of two branches within the same field do not agree on something as fundamental as the convention used to denote flow in diagrams. It is even worse when people within the field argue over which convention is best. So long as one is consistent with their convention and with their thinking, *it does not matter!* Many fine technologists may be found on either side of this "fence," and some are adept enough to switch between both without getting confused.